

Program3:

Code a Dockerized python flask or nodejs application

Program3

|
|--[app.py](#)
|--requirements.txt
|--Dockerfile

Step1: create files required

[app.py](#)

```
from flask import Flask

app=Flask(__name__)

@app.route("/")
def main():
    return "hello python world"

if __name__ == "__main__":
    app.run(host="0.0.0.0",port=5000)
```

Dockerfile

```
FROM python:3.9-slim
```

```
WORKDIR /app
```

```
COPY requirements.txt ./
```

```
RUN pip install --no-cache-dir -r requirements.txt
```

```
COPY . .
```

```
EXPOSE 5000
```

```
CMD ["python", "app.py"]
```

Step2:Build and run the dockerfile

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program3$ docker build -t program3 .
[+] Building 9.2s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 200B                             0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 1.9s
=> [auth] library/python:pull token for registry-1.docker.io    0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                   0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f1 0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 681B                                 0.0s
=> CACHED [2/5] WORKDIR /app                                    0.0s
=> [3/5] COPY requirements.txt ./                               0.0s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt     7.0s
=> [5/5] COPY . .                                              0.0s
=> exporting to image                                           0.2s
=> => exporting layers                                          0.2s
=> => writing image sha256:b02f269bafd19ea7cff8479a67a767846b4032d3f19e476c2b89f26bb94 0.0s
=> => naming to docker.io/library/program3                     0.0s
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program3$ docker run -p 5000:5000 program1
* Serving Flask app 'app.py'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a product
ion WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
```

Output

< > ↻ 📌 ⓘ localhost:5000

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Hello world